

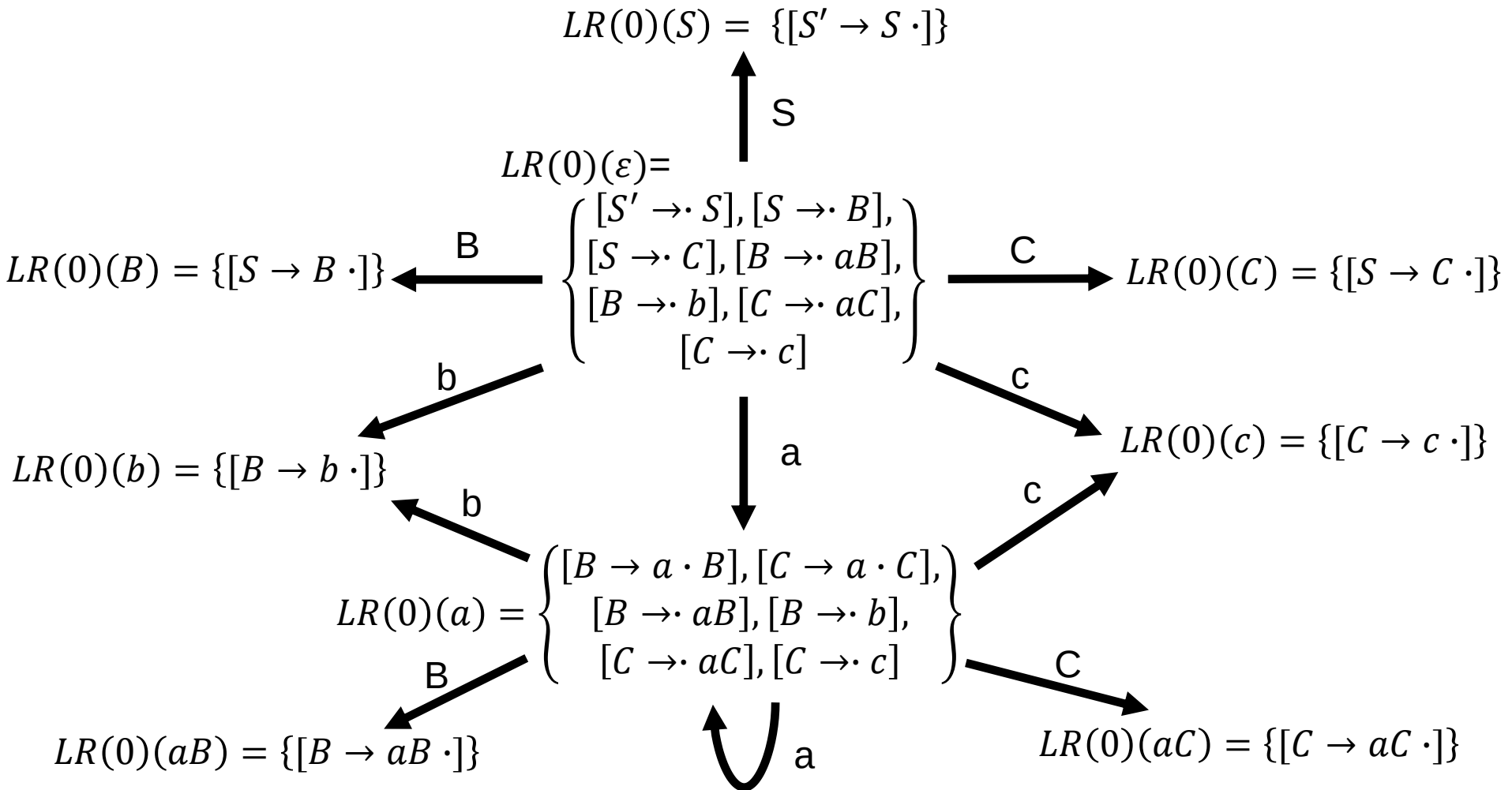
Example Parsing $LR(0)$

Recap: Deterministic Parsing Automaton $LR(0)$

- We define the deterministic parsing automaton of $LR(0)$ grammar $G = \langle \Sigma, N, P, S \rangle$ with rule 0: $S' \rightarrow S$
 - Input alphabet Σ
 - Output alphabet $U =$ the rule numbers $0, 1, 2, 3, \dots$
 - States $\Sigma^* \times \Gamma^* \times U^*$
 - Pushdown alphabet $\Gamma = LR(0)(G)$!! item sets !!
 - Initial state (w, I_0, ε) for $w \in \Sigma^*$ where $I_0 = LR(0)(\varepsilon)$
 - Final state final state $(\varepsilon, \varepsilon, u)$ where $u \in U^*$
 - Action depends on the item set I in the state $(x, \alpha I, z)$:
 - **Shift** $(aw, \alpha I, z) \rightarrow (w, \alpha IJ, z)$ if $[A \rightarrow \alpha_1 \cdot a \alpha_2] \in I$ and $I \xrightarrow[a]{\text{goto}} J$
 - **Reduce** $(aw, \alpha I I_1 \dots I_n, z) \rightarrow (w, \alpha IJ, zi)$ with rule $i \neq 0$ $A \rightarrow Y_1 \dots Y_n$ if $[A \rightarrow Y_1 \dots Y_n \cdot] \in I_n$ and $I \xrightarrow[A]{\text{goto}} J$
 - **Accepting** state $(\varepsilon, I_0 I, z) \rightarrow (\varepsilon, \varepsilon, z0)$ if $[S' \rightarrow S \cdot] \in I$
 - **Error** in state $(w, \alpha I, z)$ if $I = \emptyset$

Example, step 1: Construct the goto-automaton

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)



Example, step 2: goto-automaton as table

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

- For the parsing, it is more convenient to write the transitions of the goto automaton as a table (all empty cells are “illegal” and go to I_9)

Item set	S	B	C	a	b	c
$I_0 = LR(0)(\varepsilon)$	I_1	I_2	I_3	I_4	I_5	I_6
$I_1 = LR(0)(S)$						
$I_2 = LR(0)(B)$						
$I_3 = LR(0)(C)$						
$I_4 = LR(0)(a)$		I_7	I_8	I_4	I_5	I_6
$I_5 = LR(0)(b)$						
$I_6 = LR(0)(c)$						
$I_7 = LR(0)(aB)$						
$I_8 = LR(0)(aC)$						
$I_9 = \emptyset$						

$$I_0 = LR(0)(\varepsilon) = \left\{ \begin{array}{l} [S' \rightarrow \cdot S], [S \rightarrow \cdot B], \\ [S \rightarrow \cdot C], [B \rightarrow \cdot aB], \\ [B \rightarrow \cdot b], [C \rightarrow \cdot aC], \\ [C \rightarrow \cdot c] \end{array} \right\}$$

$$I_1 = LR(0)(S) = \{[S' \rightarrow S \cdot]\}$$

$$I_2 = LR(0)(B) = \{[S \rightarrow B \cdot]\}$$

$$I_3 = LR(0)(C) = \{[S \rightarrow C \cdot]\}$$

$$I_4 = LR(0)(a) = \left\{ \begin{array}{l} [B \rightarrow a \cdot B], [C \rightarrow a \cdot C], \\ [B \rightarrow \cdot aB], [B \rightarrow \cdot b], \\ [C \rightarrow \cdot aC], [C \rightarrow \cdot c] \end{array} \right\}$$

$$I_5 = LR(0)(b) = \{[B \rightarrow b \cdot]\}$$

$$I_6 = LR(0)(c) = \{[C \rightarrow c \cdot]\}$$

$$I_7 = LR(0)(aB) = \{[B \rightarrow aB \cdot]\}$$

$$I_8 = LR(0)(aC) = \{[C \rightarrow aC \cdot]\}$$

$$I_9 = \emptyset$$

Example, step 3: Determine actions for parser automaton

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

- For every item set I , we determine the action for the automaton to take when seeing I in a state $(w, \gamma I, z)$

Item set	action
$I_0 = LR(0)(\varepsilon)$	shift
$I_1 = LR(0)(S)$	accept
$I_2 = LR(0)(B)$	reduce 1
$I_3 = LR(0)(C)$	reduce 2
$I_4 = LR(0)(a)$	shift
$I_5 = LR(0)(b)$	reduce 4
$I_6 = LR(0)(c)$	reduce 6
$I_7 = LR(0)(aB)$	reduce 3
$I_8 = LR(0)(aC)$	reduce 5
$I_9 = \emptyset$	error

$$I_0 = LR(0)(\varepsilon) = \left\{ \begin{array}{l} [S' \rightarrow \cdot S], [S \rightarrow \cdot B], \\ [S \rightarrow \cdot C], [B \rightarrow \cdot aB], \\ [B \rightarrow \cdot b], [C \rightarrow \cdot aC], \\ [C \rightarrow \cdot c] \end{array} \right\}$$

$$I_1 = LR(0)(S) = \{[S' \rightarrow S \cdot]\}$$

$$I_2 = LR(0)(B) = \{[S \rightarrow B \cdot]\}$$

$$I_3 = LR(0)(C) = \{[S \rightarrow C \cdot]\}$$

$$I_4 = LR(0)(a) = \left\{ \begin{array}{l} [B \rightarrow a \cdot B], [C \rightarrow a \cdot C], \\ [B \rightarrow \cdot aB], [B \rightarrow \cdot b], \\ [C \rightarrow \cdot aC], [C \rightarrow \cdot c] \end{array} \right\}$$

$$I_5 = LR(0)(b) = \{[B \rightarrow b \cdot]\}$$

$$I_6 = LR(0)(c) = \{[C \rightarrow c \cdot]\}$$

$$I_7 = LR(0)(aB) = \{[B \rightarrow aB \cdot]\}$$

$$I_8 = LR(0)(aC) = \{[C \rightarrow aC \cdot]\}$$

$$I_9 = \emptyset$$

Run the parser automaton, part 1

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Input aac

■ Parsing:

- The initial state is (aac, I_0, ε) with $I_0 = LR(0)(\varepsilon)$
- What do we have to do next? Let's look at the table:

Item set	action
I_0	shift

- We have to shift, taking the first “a” from the input
- What will happen next? Consult the table:

Item set	S	B	C	a	b	c
I_0	I_1	I_2	I_3	I_4	I_5	I_6

- We go now to I_4 . The new state is $(ac, I_0 I_4, \varepsilon)$
- Note that I_0 is not removed. We are not yet done with it!

Run the parser automaton, part 2

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Parsing:

- The current state is $(ac, I_0 I_4, \varepsilon)$
- What do we have to do next? Let's look at the table:

Item set	action
I_4	shift

Item set	S	B	C	a	b	c
I_4		I_7	I_8	I_4	I_5	I_6

- The new state is $(c, I_0 I_4 I_4, \varepsilon)$

Run the parser automaton, part 3

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Parsing:

- The current state is $(c, I_0 I_4 I_4, \varepsilon)$
- What do we have to do next? Let's look at the table:

Item set	action
I_4	shift

Item set	S	B	C	a	b	c
I_4		I_7	I_8	I_4	I_5	I_6

- The new state is $(\varepsilon, I_0 I_4 I_4 I_6, \varepsilon)$

Run the parser automaton, part 4

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Parsing:

- The current state is $(\varepsilon, I_0 I_4 I_4 I_6, \varepsilon)$
- What do we have to do next? Let's look at the table:

Item set	action
I_6	reduce 6

- That means we are ready to reduce to C using rule 6: $C \rightarrow c$
- The right-hand side of $C \rightarrow c$ contains one symbol, so we remove one element from the stack:

$(\varepsilon, I_0 I_4 I_4, 6)$

And we consult the table to see where $I_4 \xrightarrow{C}$ leads us:

Item set	S	B	C	a	b	c
I_4		I_7	I_8	I_4	I_5	I_6

- The new state is $(\varepsilon, I_0 I_4 I_4 I_8, 6)$

Run the parser automaton, part 5

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Parsing:

- The current state is $(\epsilon, I_0 I_4 I_4 I_8, 6)$
- What do we have to do next? Let's look at the table:

Item set	action
I_8	reduce 5

- That means we are ready to reduce to C using rule 5: $C \rightarrow aC$
- The right-hand side of $C \rightarrow aC$ contains two symbols, so we remove two elements from the stack:

$(\epsilon, I_0 I_4, 65)$

And we consult the table to see where $I_4 \xrightarrow{C}$ leads us:

Item set	S	B	C	a	b	c
I_4		I_7	I_8	I_4	I_5	I_6

- The new state is $(\epsilon, I_0 I_4 I_8, 65)$

Run the parser automaton, part 6

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Parsing:

- The current state is $(\epsilon, I_0 I_4 I_8, 65)$
- What do we have to do next? Let's look at the table:

Item set	action
I_8	reduce 5

- That means we are ready to reduce to C using rule 5: $C \rightarrow aC$
- The right-hand side of $C \rightarrow aC$ contains two symbols, so we remove two elements from the stack:

$(\epsilon, I_0, 655)$

And we consult the table to see where $I_0 \xrightarrow{C}$ leads us:

Item set	S	B	C	a	b	c
I_0	I_1	I_2	I_3	I_4	I_5	I_6

- The new state is $(\epsilon, I_0 I_3, 655)$

Run the parser automaton, part 7

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Parsing:

- The current state is $(\epsilon, I_0 I_3, 655)$
- What do we have to do next? Let's look at the table:

Item set	action
I_3	reduce 2

- That means we are ready to reduce to S using rule 2: $S \rightarrow C$
- The right-hand side of $S \rightarrow C$ contains one symbol, so we remove one element from the stack:

$(\epsilon, I_0, 6552)$

And we consult the table to see where $I_0 \xrightarrow{S}$ leads us:

Item set	S	B	C	a	b	c
I_0	I_1	I_2	I_3	I_4	I_5	I_6

- The new state is $(\epsilon, I_0 I_1, 6552)$

Run the parser automaton, part 8

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Parsing:

- The current state is $(\varepsilon, I_0 I_1, 6552)$
- What do we have to do next? Let's look at the table:

Item set	action
I_1	accept

- We are done. The last state is $(\varepsilon, \varepsilon, 65520)$

Everything in one picture

Example: $S' \rightarrow S$ (0) $S \rightarrow B|C$ (1,2)
 $B \rightarrow aB|b$ (3,4) $C \rightarrow aC|c$ (5,6)

■ Input aac

■ Parsing:

(aac, I_0, ε)
 $(ac, I_0I_4, \varepsilon)$
 $(c, I_0I_4I_4, \varepsilon)$
 $(\varepsilon, I_0I_4I_4I_6, \varepsilon)$
 $(\varepsilon, I_0I_4I_4I_8, 6)$
 $(\varepsilon, I_0I_4I_8, 65)$
 $(\varepsilon, I_0I_3, 655)$
 $(\varepsilon, I_0I_1, 6552)$
 $(\varepsilon, \varepsilon, 65520)$

Item set	action	S	B	C	a	b	c
$I_0 = LR(0)(\varepsilon)$	shift	I_1	I_2	I_3	I_4	I_5	I_6
$I_1 = LR(0)(S)$	accept						
$I_2 = LR(0)(B)$	red 1						
$I_3 = LR(0)(C)$	red 2						
$I_4 = LR(0)(a)$	shift		I_7	I_8	I_4	I_5	I_6
$I_5 = LR(0)(b)$	red 4						
$I_6 = LR(0)(c)$	red 6						
$I_7 = LR(0)(aB)$	red 3						
$I_8 = LR(0)(aC)$	red 5						
$I_9 = \emptyset$	error						